Ideation Phase

Define the Problem Statements

Date	2 March 2026
Team ID	
Project Name	PlantCare AI: Advanced Plant Disease Detection Using Transfer Learning
Maximum Marks	2 Marks

Customer Problem Statement Template:

In agriculture, early detection of plant diseases is very important to prevent crop loss and improve productivity. Many farmers face difficulty in identifying plant diseases at an early stage because they rely on manual observation of plant leaves. Due to the lack of agricultural experts and modern diagnostic tools, farmers may not correctly identify the disease affecting their crops.

Understanding the farmer's perspective is essential to develop an effective solution. Farmers need a simple and reliable method to detect plant diseases quickly so that they can take preventive measures and protect their crops.

To address this problem, our project **PlantCare AI** proposes an artificial intelligence based system that detects plant diseases using leaf images and provides early diagnosis. This system will help farmers identify diseases accurately and reduce crop damage.

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A farmer	identify plant diseases in crops early	I cannot easily recognize diseases on plant leaves	I do not have access to agricultural experts or tools	worried about crop damage and financial loss

PS-2	A farmer	maintain healthy crops and improve productivity	I cannot detect plant diseases quickly	manual inspection is difficult and time consuming	frustrated and uncertain about crop yield